

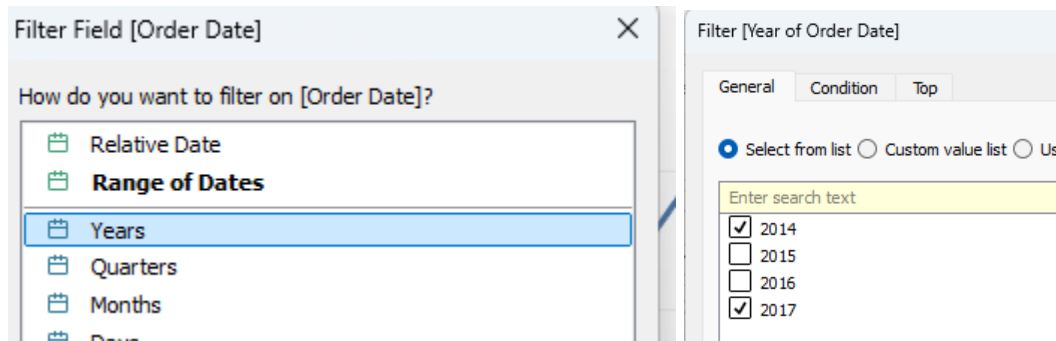
Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – XI

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Class:IT	Batch:1
Date of Assignment:01-09-2026	Date/Time of Submission:01-09-2026

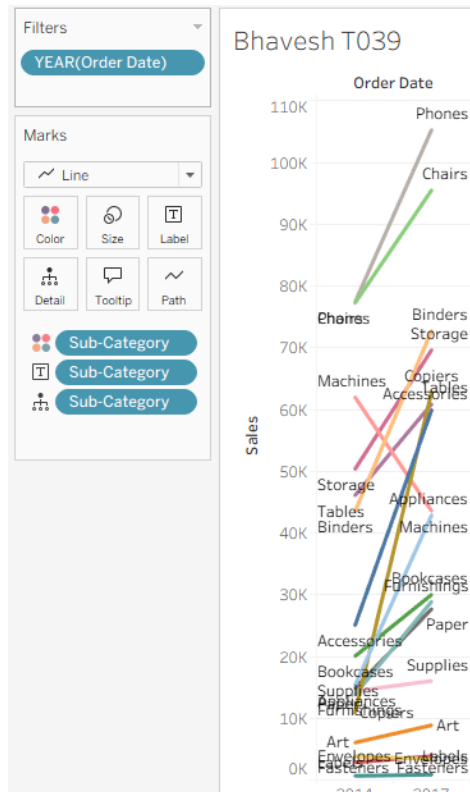
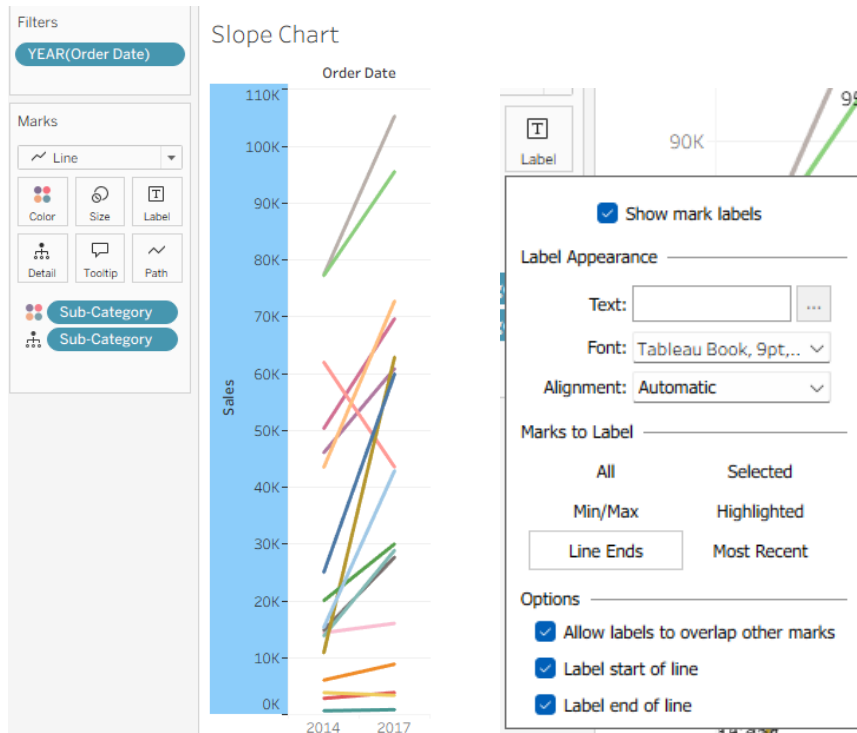
A: Create Slope Charts and Lollipop Charts

1. Create a Slope Chart (Year-over-Year Progression)

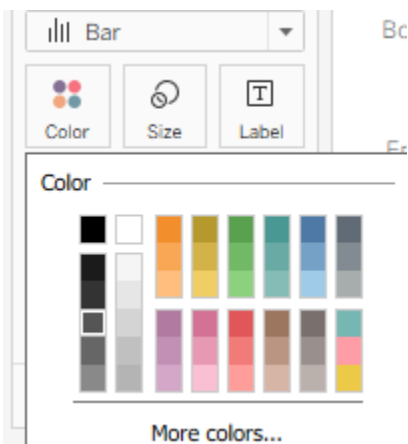
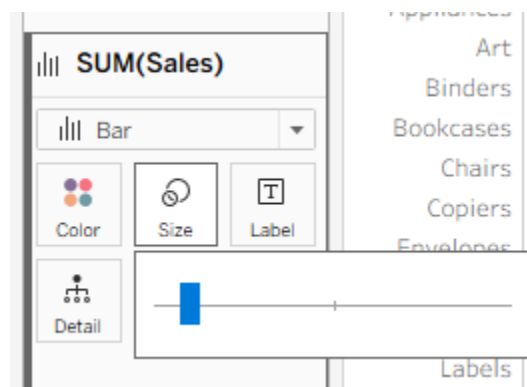
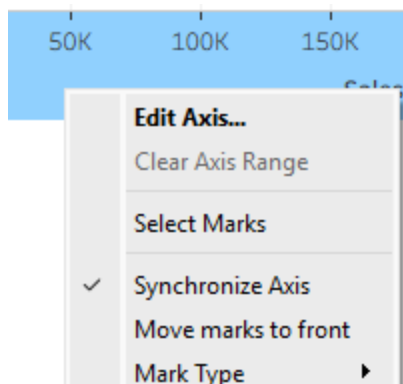
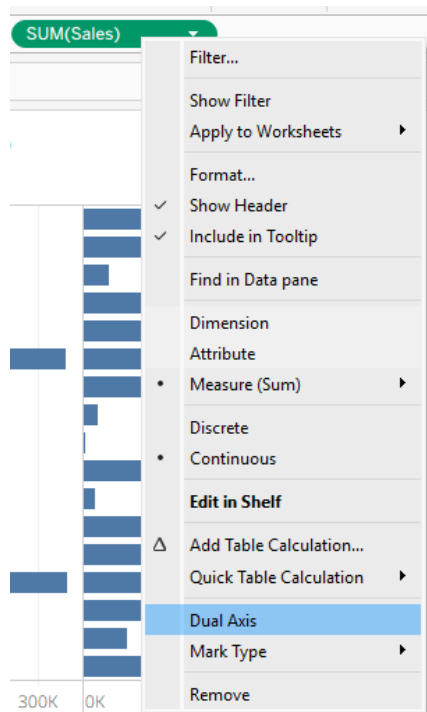


3. Right-click on the YEAR(Order Date) pill on the Columns shelf --> select Discrete (blue pill) so it acts as two distinct category pillars.

Step 3: Disaggregate by Sub-Category



2. Create a Lollipop Chart (Sales with Profit Highlighting)



Marks

All

SUM(Sales)

SUM(Sales) (2)

Circle

Color

Size

Label

Detail

Sub-Catego..

Accessories

Appliances

Art

Binders

Bookcases

Chairs

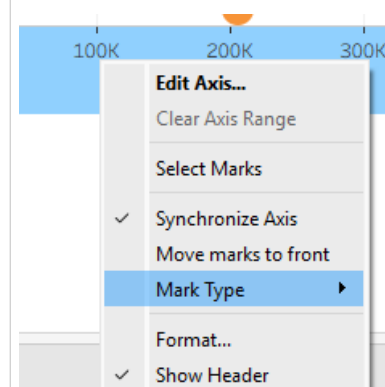
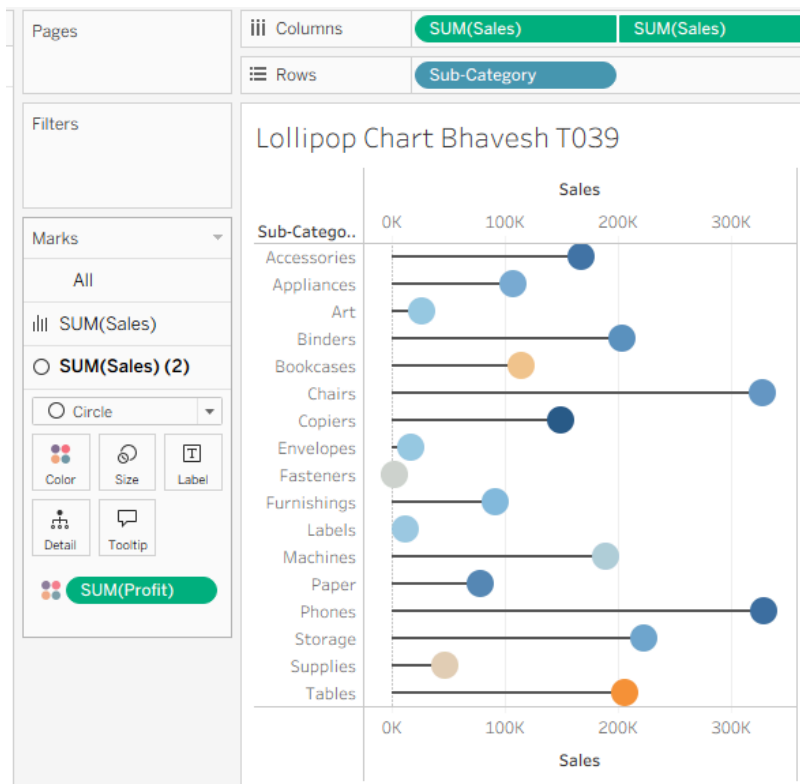
Copiers

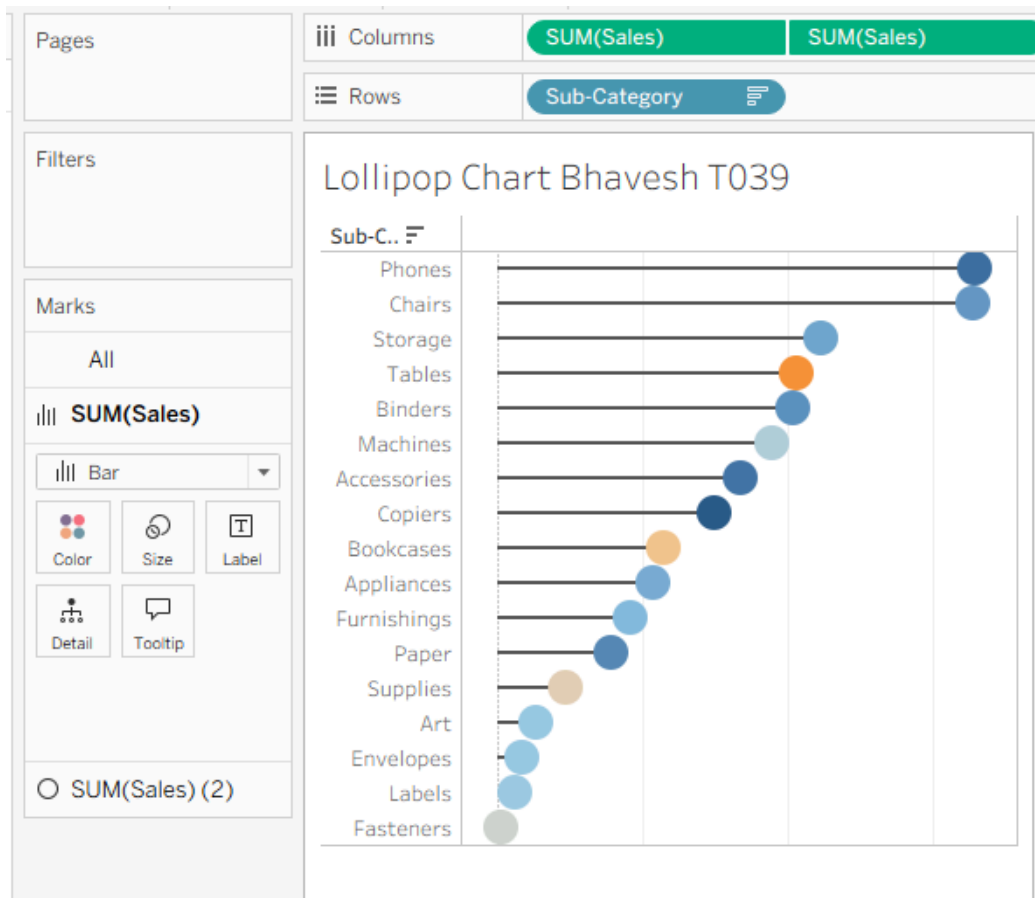
Envelopes

Fasteners

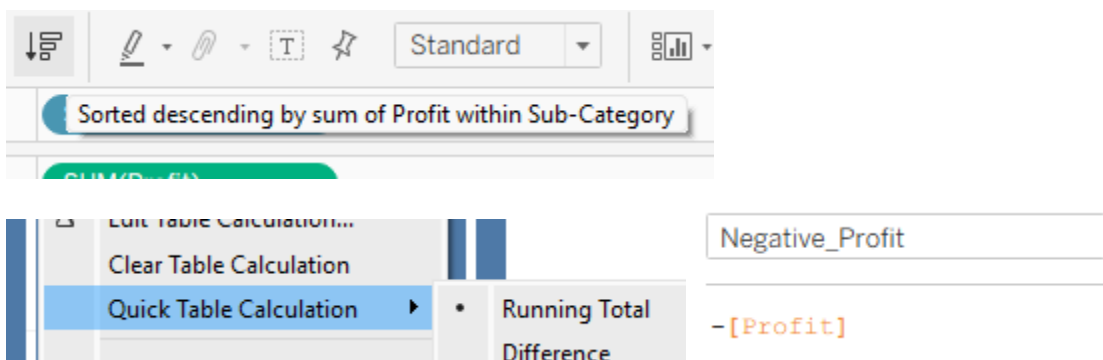
Paper

Phones





B: Design Waterfall Charts



Edit Colors SUM(Profit)

Palette

Green-Blue Diverging



Start

Automatic -17,725.4811

Center

Automatic 0

End

Automatic 55,617.8249

☐ Stepped Color 5 steps

☐ Reverse

☒ Use full color range

☐ Include totals

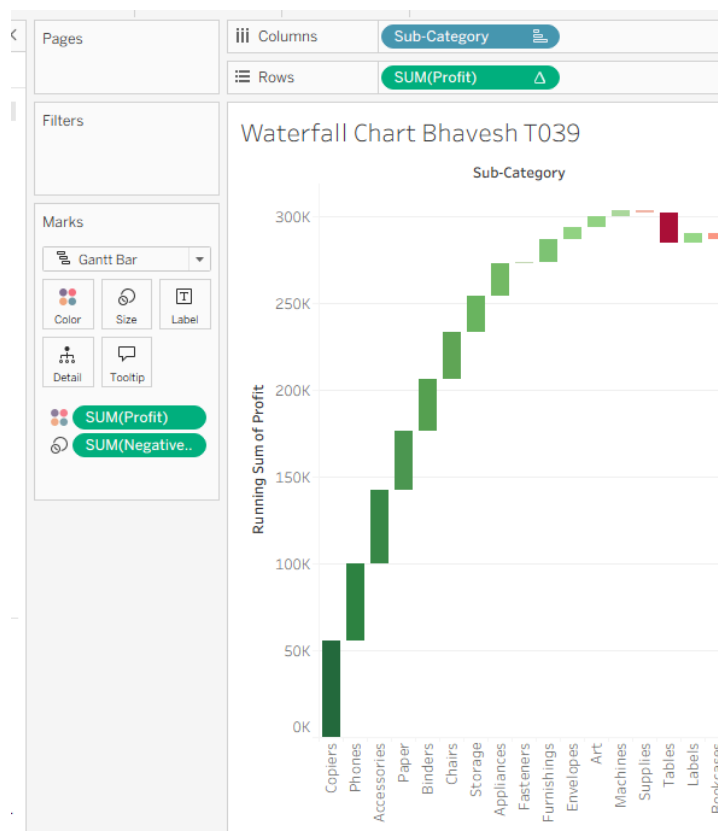
Color Size Label

Color

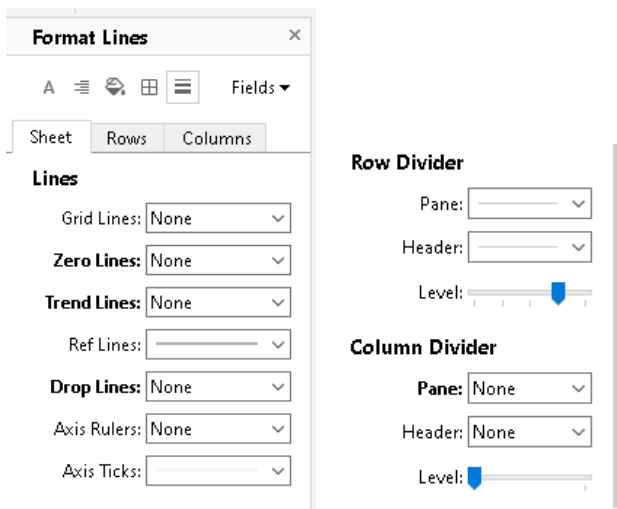
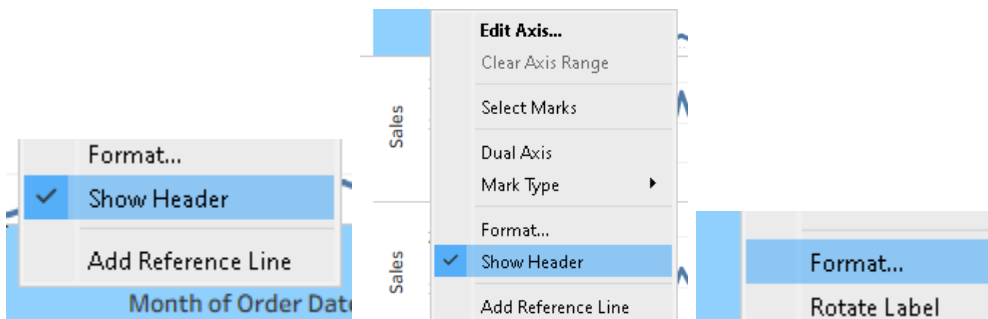
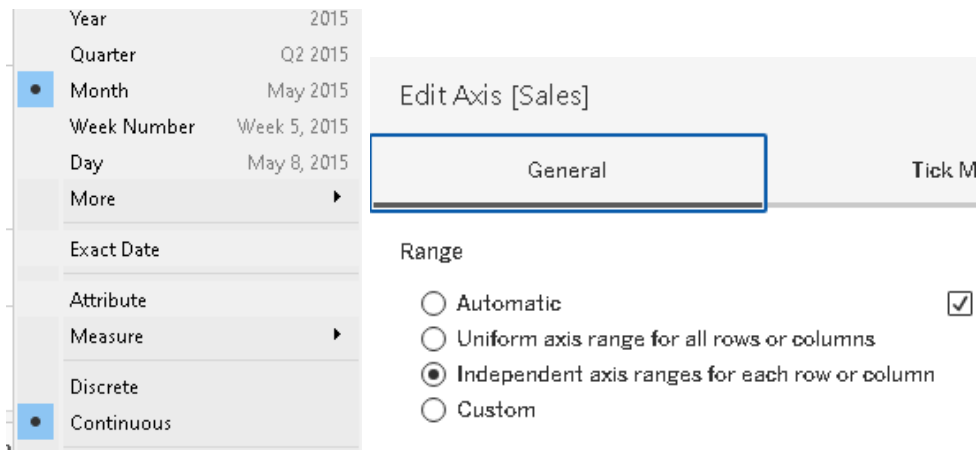
Edit Colors...

Opacity

Percentage Of	Category
Totals	Show Row Grand Totals
Forecast	Show Column Grand Totals



C: Build Sparklines for Trend Analysis



Step 5: Highlight Minimum and Maximum Milestone Endpoints

☒ Show mark labels

Label Appearance

Text: ...

Font: ▼

Alignment: ▼

Marks to Label

All Selected

Min/Max Highlighted

Line Ends Most Recent

Scope

Table Cell

Pane Line/Pie

Field

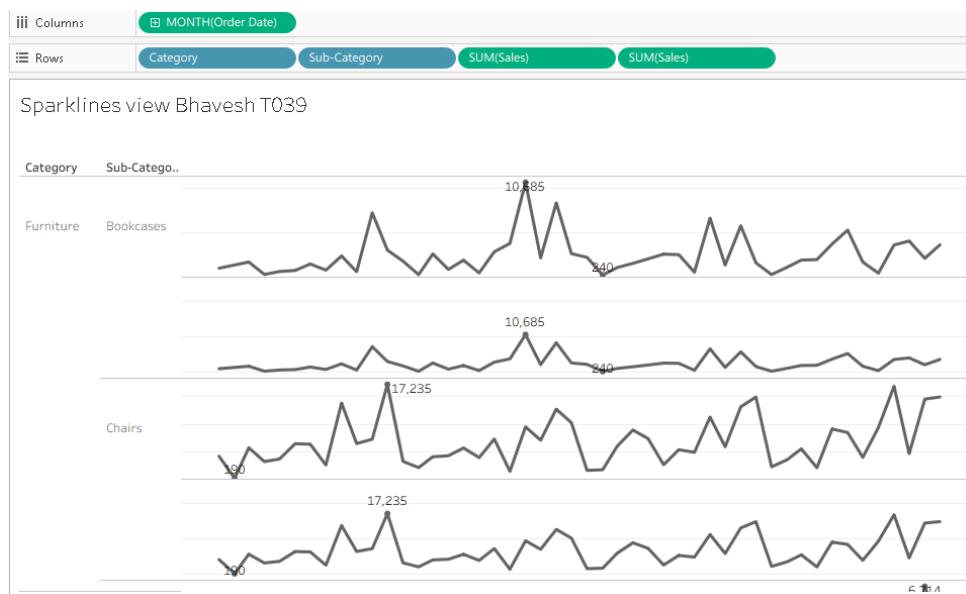
▼

Options

☒ Allow labels to overlap other marks

☒ Label minimum value

☒ Label maximum value



D: Implement Sheet Swapping for Dynamic Dashboards

Create Calculated Field...

Create Parameter...

Create Folder (use group by folder)

Create Parameter

Name

Select_View_Parameter

Properties

Data type

String

Display format

Category Sales(bar)

Current value

Category Sales(bar)

Value when workbook opens

Current value

Allowable values

☐ All

☒ List

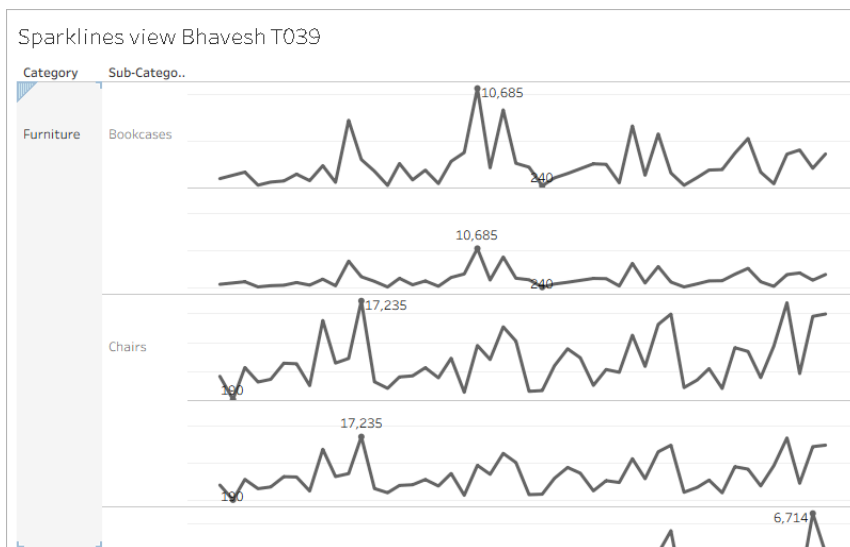
☐ Range

Value	Display As
Category Sales(bar)	Category Sales(bar)
Regional Profit (Map)	Regional Profit (Map)
Click to add	

☒ Fixed

☐ When workbook opens

Add values from



Step 2: Create the Dynamic Filter Field

Filter [View_Filter]

General Wildcard Condition Top

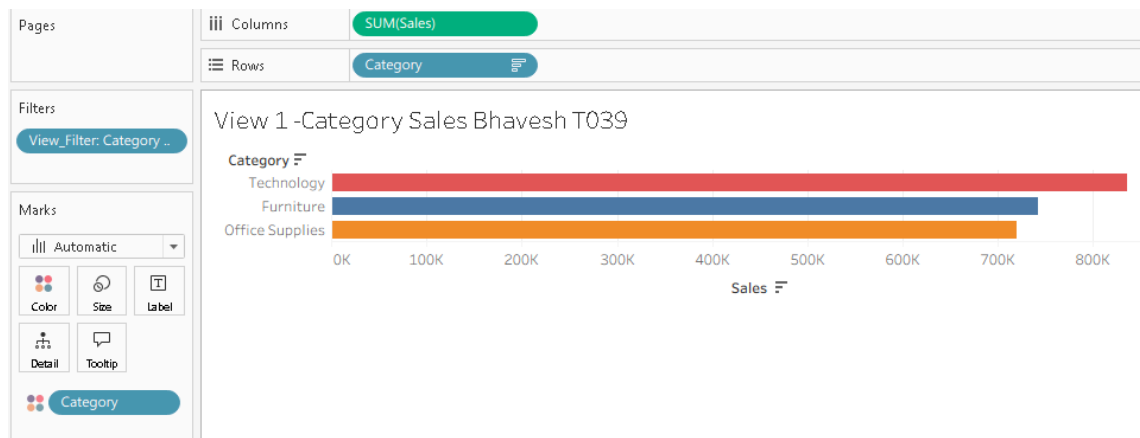
☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

☒ Category Sales(bar)

View_Filter

[Select_View_Parameter]



Step 4: Build Sheet 2 (View 2 - Regional Profit)

Filter [View_Filter]

General Wildcard Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

☒ Regional Profit (Map)

Select_View_Parameter

Regional Profit (Map) ▼

View 2 - Regional Profit

Edit Title...

Reset Title

Hide Title

Format Title...

Category

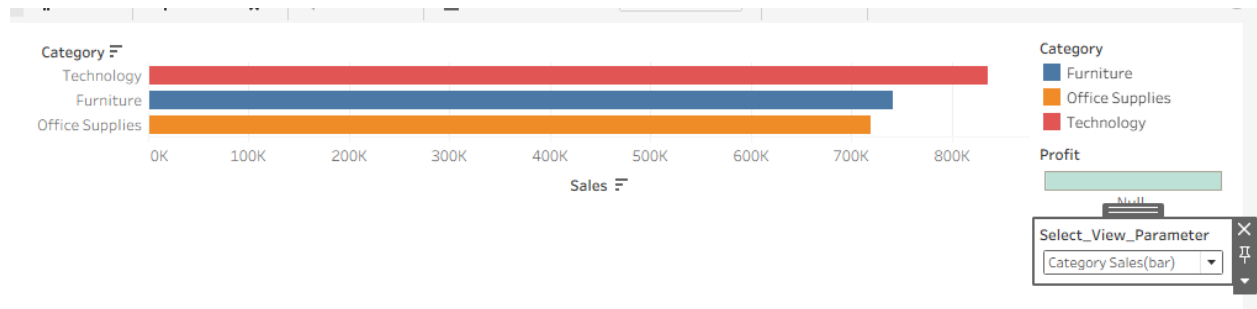
Profit



Select_View_Parameter

Regional Profit (Map) ▼





E: Create Custom Geographic Territories

City

Country

Customer

Customer

Order Date

Order ID

Postal Code

Product ID

Product ID

Region

Row ID

Segment

Ship Date

Ship Mode

State

Create

Transform

Convert to Measure

Change Data Type

Default Properties

Geographic Role

Image Role

Group by

Folders

Hierarchy

Replace References...

Describe...

Calculated Field...

Group...

Set...

Parameter...

Size

Text

Tooltip

Alabama

Pacific & Mountain Zone

Arizona

California

Colorado

Idaho

Montana

Nevada

New Mexico

Oregon

Utah

Washington

Wyoming

Georgia	
Midwest & Central Hub	
Illinois	
Indiana	
Iowa	
Kansas	
Michigan	
Minnesota	
Missouri	
Nebraska	
North Dakota	
Ohio	
Oklahoma	
South Dakota	
Texas	
Wisconsin	
	Atlantic & North-East
	Connecticut
	Delaware
	Maine
	Maryland
	Massachusetts
	New Hampshire
	New Jersey
	New York
	Pennsylvania
	Rhode Island
	Vermont

Create Group [State] X

Field Name: Sales_Territories

Groups: Add to:

Southern Enterprise Territory

Alabama

Arkansas

Delaware

Florida

Georgia

Kentucky

Louisiana

Mississippi

North Carolina

South Carolina

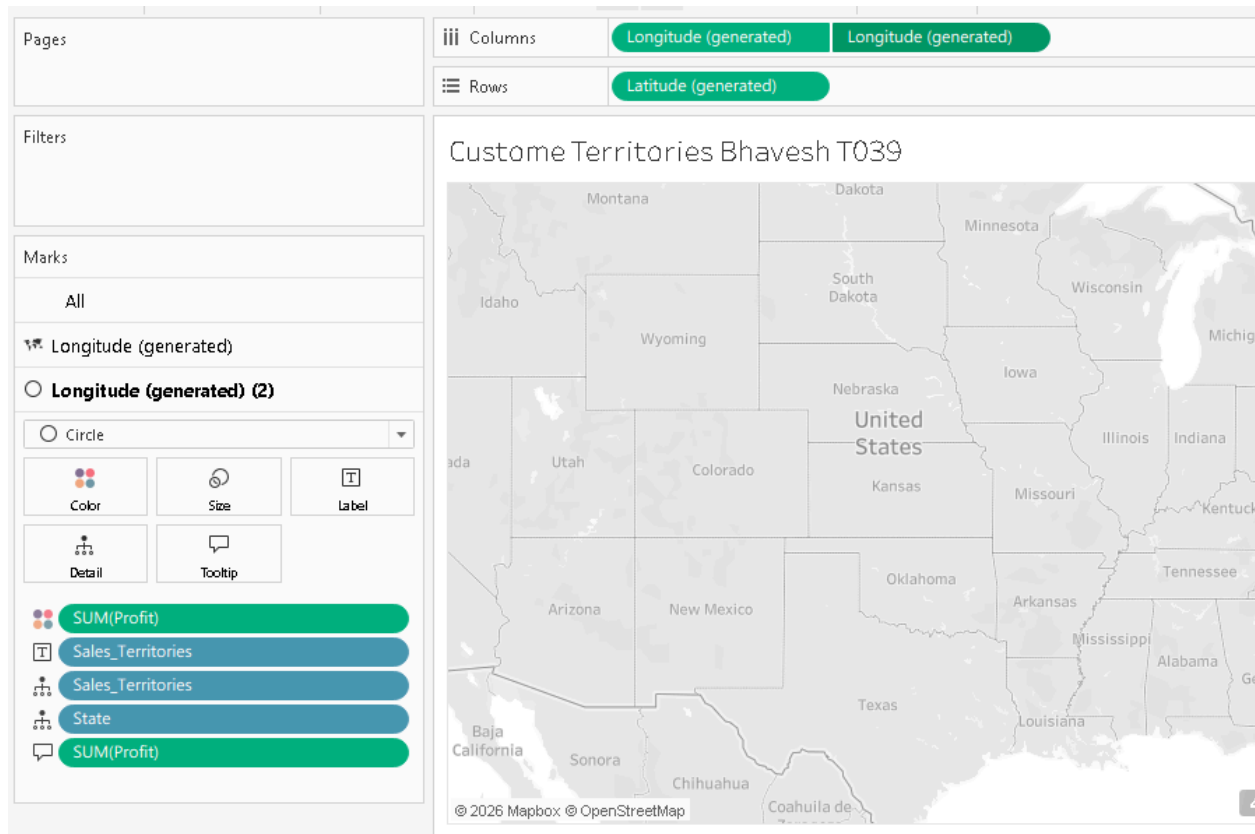
Tennessee

Virginia

Pacific & Mountain Zone

Step 2: Assign Geographic Role to the Custom Group

Step 3: Build the Consolidated Territory Map



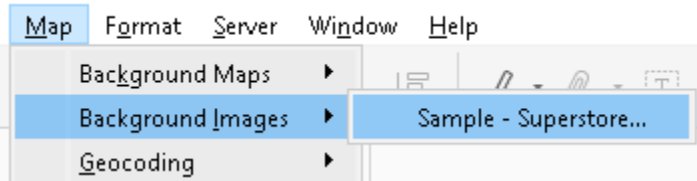
F: Use Background Images in Dashboards

Store_X_Coordinate

```
CASE [Sub-Category]
WHEN "Phones" THEN 150
WHEN "Accessories" THEN 250
WHEN "Binders" THEN 150
WHEN "Chairs" THEN 500
WHEN "Tables" THEN 500
WHEN "Bookcases" THEN 380
WHEN "Storage" THEN 800
WHEN "Appliances" THEN 680
WHEN "Machines" THEN 850
WHEN "Copiers" THEN 850
ELSE 500
END
```

Store_Y_Coordinate

```
CASE [Sub-Category]
WHEN "Phones" THEN 480
WHEN "Accessories" THEN 360
WHEN "Binders" THEN 200
WHEN "Chairs" THEN 480
WHEN "Tables" THEN 200
WHEN "Bookcases" THEN 100
WHEN "Storage" THEN 480
WHEN "Appliances" THEN 100
WHEN "Machines" THEN 200
WHEN "Copiers" THEN 360
ELSE 300
END
```



Add Background Image

Name: Retail Floor Plan

Image Options

File or URL:

X Field

Left:

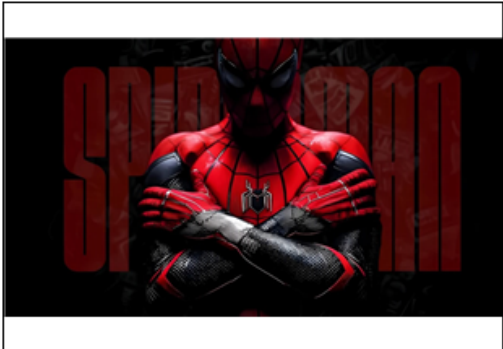
Right:

Y Field

Bottom:

Top:

Washout:



Add Background Image

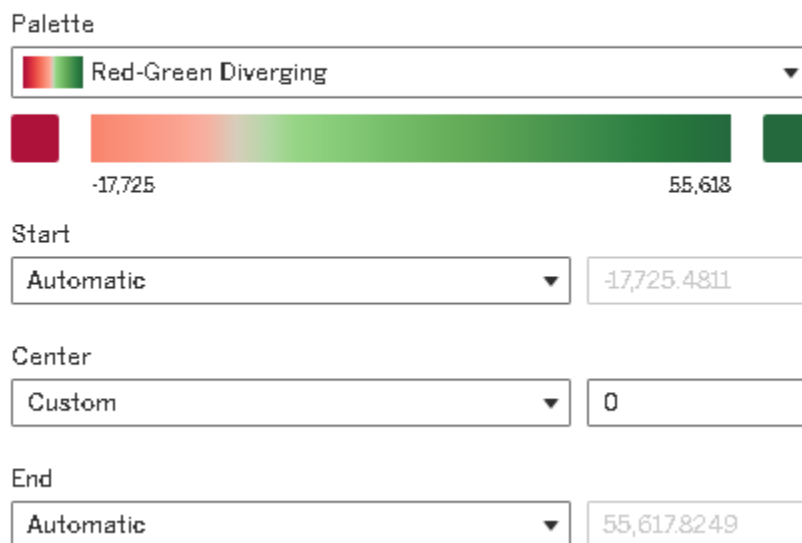
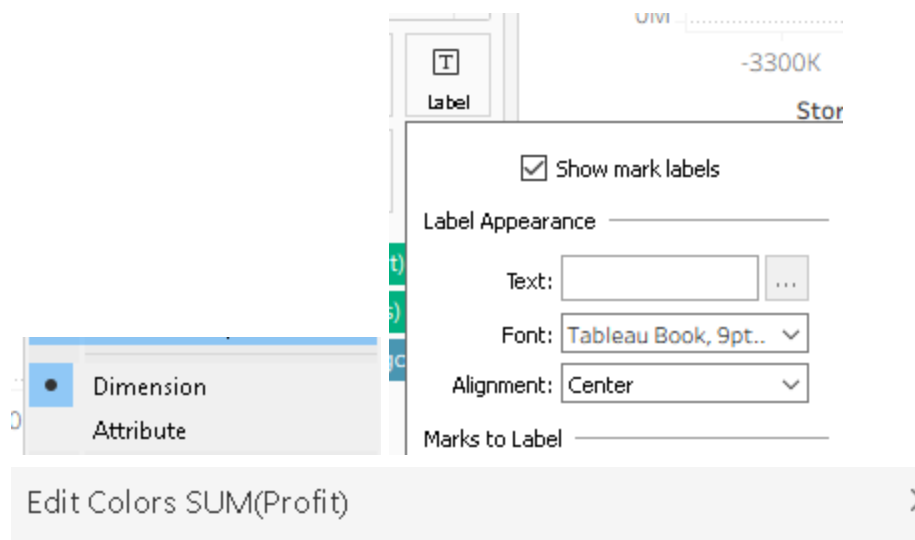
Name: Retail Floor Plan

Image Options

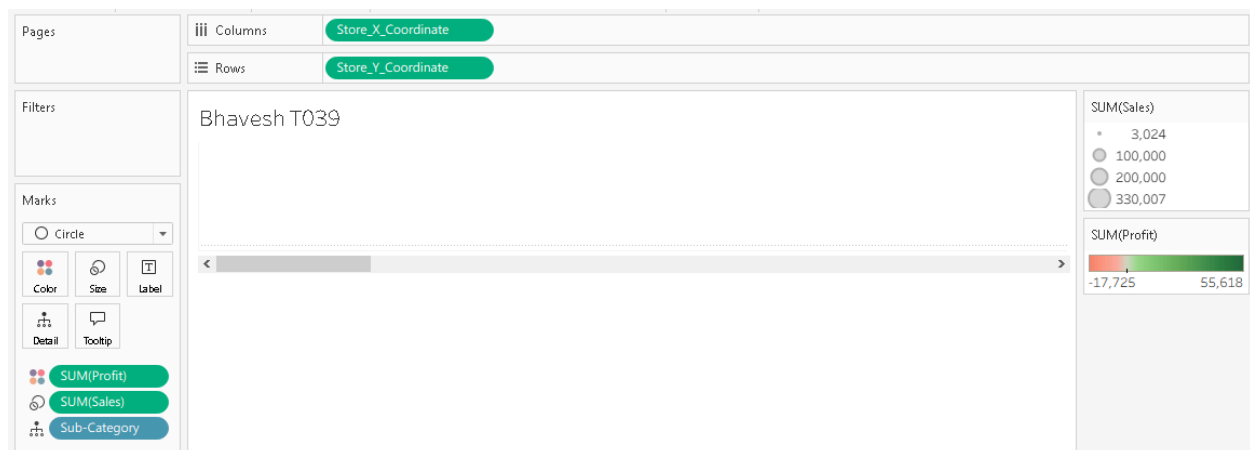
☒ Lock Aspect Ratio

☒ Always Show Entire Image

☐ Only Show When

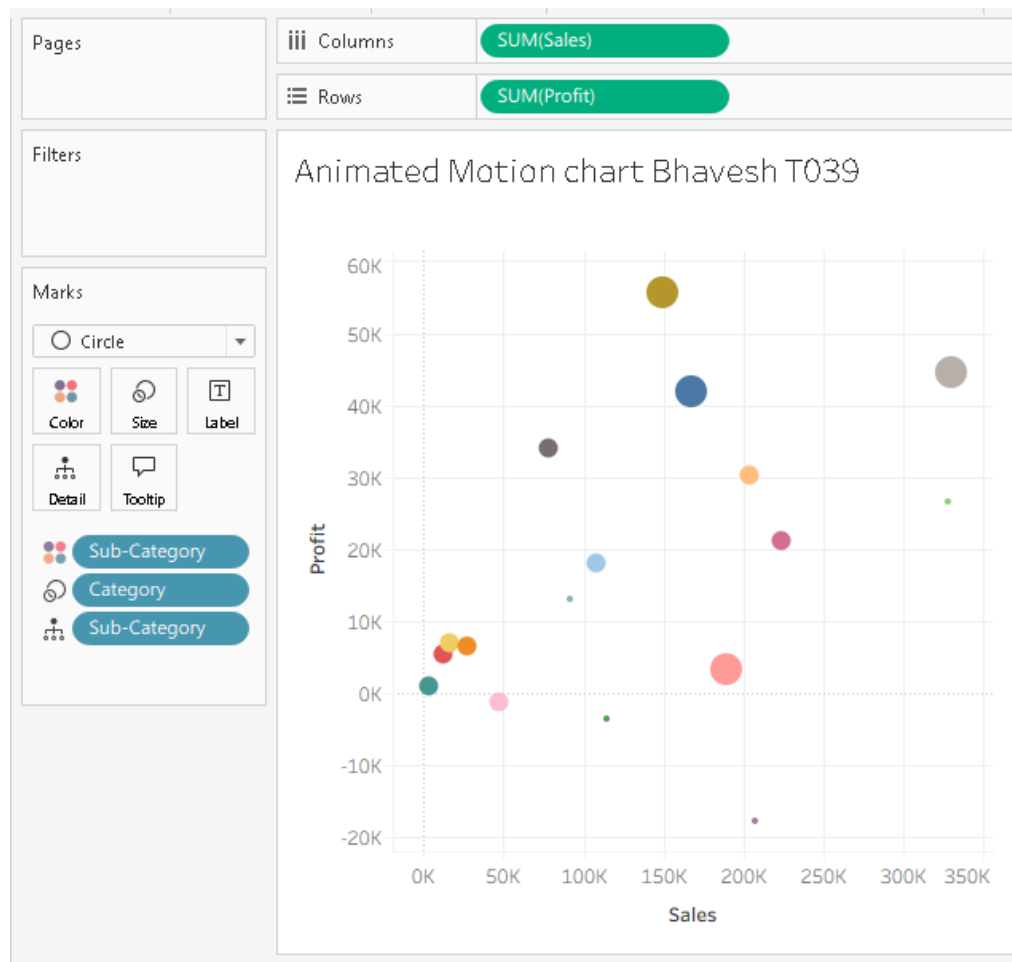


Right-click the X-axis (Store_X_Coordinate) --> uncheck Show Header.

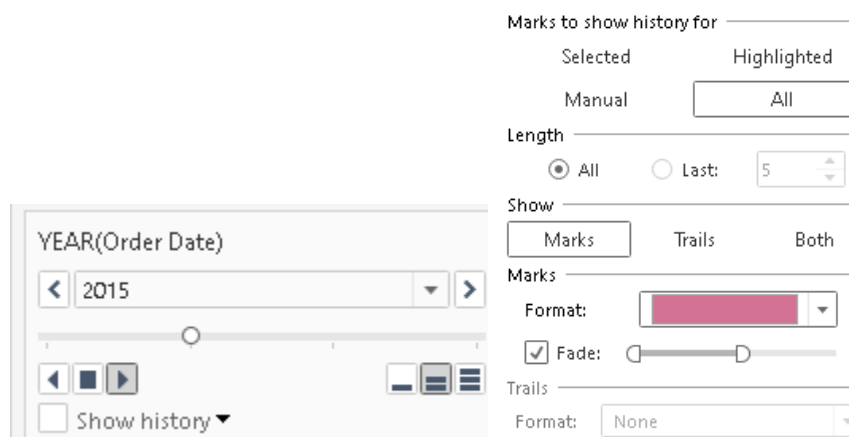


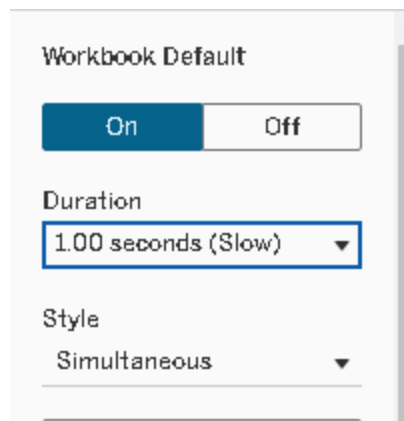
G: Apply Animation and Transparency Techniques

Task 1: Create an Animated Motion Chart (Pages Shelf)

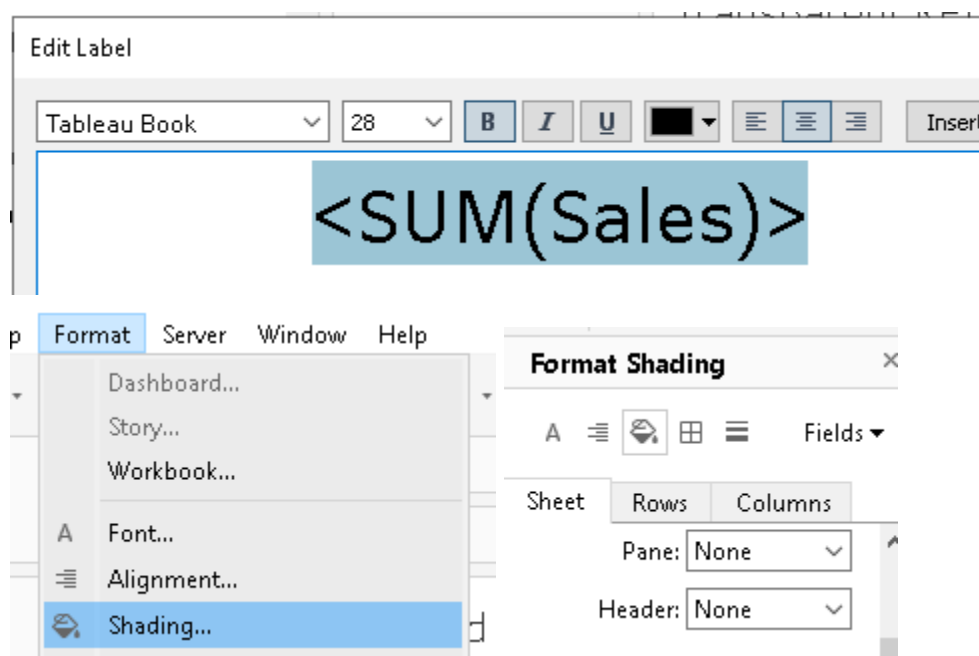


2. Right-click the Order Date pill on the Pages shelf --> select Year (discrete YEAR(Order Date)). Tableau may have selected it by default





Task 2: Apply Sheet and Dashboard Transparency



Format Dashboard

Dashboard Shading

Default:

Dashboard T

None

Font:

Alignment:

Shading:

Border:

More colors...

